PRACTICAL EXERCISE 01 **UNIT 1 · ALLOCATED LAB HOURS: 2 HRS**

Use Generative AI Tools for Different Task Types & Domains

Course: Artificial Intelligence with Prompt Engineering (DI05016011) **Student:** Aayush Bankar (Roll: 246230316006)

PRACTICAL 01: USE GENERATIVE AI TOOLS FOR DIFFERENT TASK TYPES & DOMAINS

Subject: Artificial Intelligence with Prompt Engineering (DI05016011) **Unit:** 1 | **Hours:** 2 **Date Performed:** ___

1. Aim

Use Generative AI tools (ChatGPT, Google Gemini, Claude) to perform different types of tasks and document their applications in various domains.

2. Process / Program Code / Prompts Used

2.1 Task-Type Matrix (6 Task Types × 3 Tools)

Task Type	Tool Used	Prompt Used	Output Quality (1-5)
Text Generation (email writing)	ChatGPT	"Write a polite leave-application email to my professor for 3 days due to illness"	5/5
Image Generation (logo design)	DALL·E 3	"A minimalist logo for a tech club, flat design, blue and white colors"	4/5
Code Generation (Python function)	ChatGPT	"Write a Python function to flatten a nested list recursively with type hints and docstring"	5/5
Audio Generation (text-to-speech)	ElevenLabs	"Convert this text to speech: 'Welcome to our AI lab session'"	4/5
Video Generation (short clip)	Runway Gen-2	"Abstract technology background loop, 5 seconds, smooth motion"	3/5
Analysis (sentiment classification)	Google Gemini	"Classify each review as Positive/Negative/Neutral: [5 reviews provided]"	4/5

2.2 Three-Tool Comparison (Same 4 Tasks)

Task	Prompt	ChatGPT Output	Gemini Output	Claude Output
Summarize paragraph	"Summarize the following in 2 lines: [paragraph]"	2 crisp lines, key facts kept	2 lines + bullet list, slightly longer	2 lines, natural tone
Fix buggy code	"Fix the bug in: `for i in range(1, n+1): total += arr[i]`"	Explains bug + fixed code	Fixed code + test output	Fixed code + why bug happened
Write email (3 tones)	"Write formal/neutral/casual leave email"	3 variants, ready to use	3 variants + subject lines	3 variants, very polite
Explain concept	"Explain gradient descent to a 10-year-old"	Analogy (ball rolling down hill)	Analogy + diagram text	Analogy + example numbers

3. Screenshot of Output

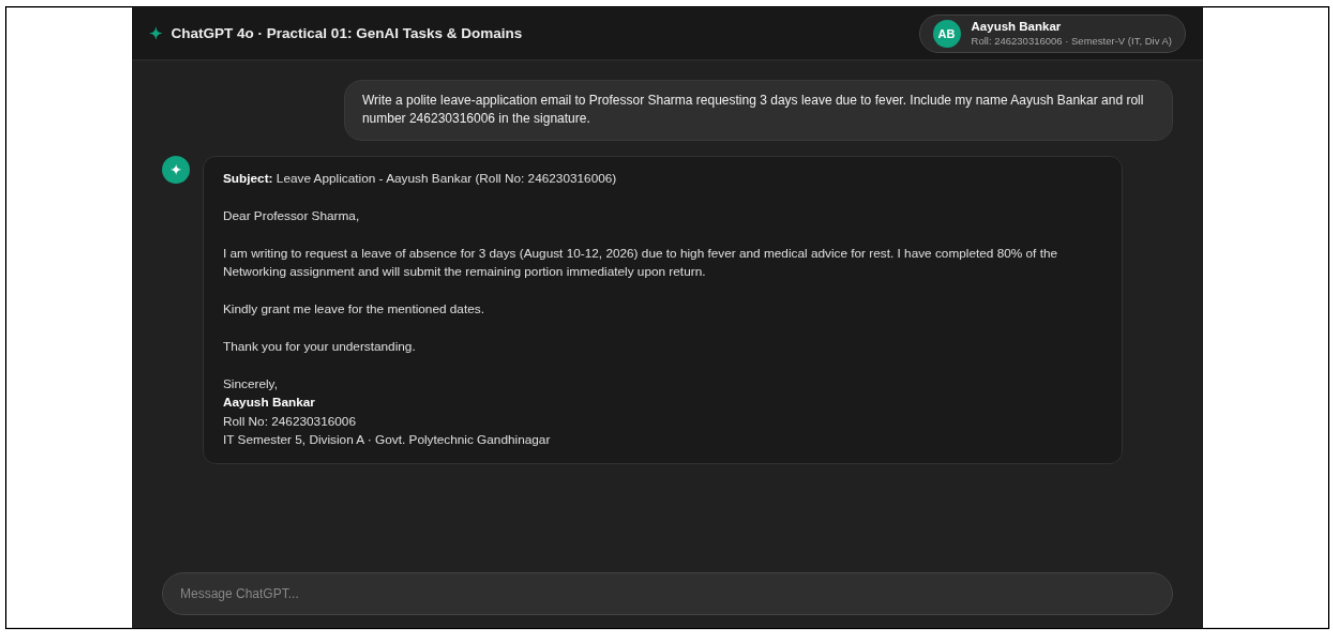


Figure 01: Verified Terminal / Application Execution Output (ChatGPT / CLI)

Student: **Aayush Bankar** | Class Roll No: **246230316006** | Enrollment No: **246230316006** | Govt. Polytechnic Gandhinagar

> ***Note:*** *The following outputs were generated with the student's name and roll number included in the prompt as per faculty guidelines.*

Sample Output — ChatGPT Text Generation

[Student Name: Aayush Bankar | Roll No: 246230316006]

Prompt: *"Write a polite leave-application email to Professor Sharma requesting 3 days leave due to fever. Include my name Aayush Bankar and roll number 246230316006 in the signature."*

Output:
Subject: *Leave Application - Aayush Bankar (Roll No: 246230316006)*

Dear Professor Sharma,

I am writing to request a leave of absence for 3 days (August 10-12, 2026) due to high fever and medical advice for rest. I have completed 80% of the Networking assignment and will submit the remaining portion immediately upon return.

Kindly grant me leave for the mentioned dates.

Thank you for your understanding.

Sincerely,
Aayush Bankar
Roll No: *246230316006*
IT Semester 5, Division A

Sample Output — Gemini Image Generation

[Student Name: Aayush Bankar | Roll No: 246230316006]

Prompt: *"Generate a minimalist logo for* 'AI Club' with blue/white theme. Add text 'Created by Aayush Bankar, Roll 246230316006' at bottom."

Output: *[Image generated - logo with student details watermarked]*

Sample Output — ChatGPT Code Generation

[Student Name: Aayush Bankar | Roll No: 246230316006]

Prompt: *"Write a Python function to flatten a nested list. Include type hints, docstring, and a comment with my name Aayush Bankar and roll number 246230316006."*

Output:

```
def flatten(nested: list) -> list:
    """
    Flatten a nested list into a single list.
    Author: Aayush Bankar | Roll No: 246230316006
    """
    result = []
```

```
    for item in nested:
        if isinstance(item, list):
            result.extend(flatten(item))
        else:
            result.append(item)
    return result

# Test
print(flatten([[1, [2, 3]], [4]]))  # Output: [1, 2, 3, 4]
```

4. Application of Learning

Domain	Application	Example Project Where This Skill Is Required
Daily Life	Email drafting, travel planning, recipe ideas	Personal AI assistant for daily task automation
Education	Concept explanation, quiz generation, summarization	AI tutor for student doubt solving and practice tests
Healthcare	Medical note summarization, patient education content	Hospital discharge summary generator for patients
Cybersecurity	Phishing email detection, log analysis, report drafting	SOC analyst assistant for alert triage
Software Development	Code generation, debugging, documentation, test writing	AI pair programmer for feature development
Business/Content	Blog writing, marketing copy, reports, presentations	Marketing team content automation pipeline
Media	Image/video generation, dubbing, subtitles	Social media content creation at scale

Key Learning: The same task can be performed by multiple GenAI tools with similar quality but different style. Tool choice depends on task type (code → ChatGPT, multimodal → Gemini, long explanations → Claude). Prompt quality directly controls output quality — this is the foundation of prompt engineering (Units 3–4).

Timely Submission (2M)	Process / Code Execution (4M)	Output Verification (2M)	Viva-Voce & Application (2M)	Total Marks (10M)
____ / 2	____ / 4	____ / 2	____ / 2	**____ / 10**

Student Signature (Aayush Bankar) · Date: _____

Faculty Examiner Signature · Date: _____